

Exercise 01: Summation

Write a text file named ‘`sums.txt`’ that finds the solution of each summation below. You need to show work to receive credit.

Summations

1. $\sum_{i=1}^{200} i$

2. $\sum_{i=10}^{80} i$

3. $\sum_{i=1}^{15} 3i^2$

4. $\sum_{i=1}^n (2i - 1)$

5. $\sum_{i=1}^n i(i + 1)$

6. $\sum_{i=1}^n (i + 1)(2i + 1)$

7. $\sum_{i=0}^{n-1} (n - i - 1)$

8. $\sum_{i=1}^n \sum_{j=i}^n j$

Extra Credit

Prove that

$$\sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}$$

given that

$$\begin{aligned} \sum_{i=1}^n (i+1)^4 - i^4 &= (n+1)^4 - 1 \\ &= n^4 + 4n^3 + 6n^2 + 4n \end{aligned}$$